

Stereoselective Synthesis of Polyfunctional Tetrasubstituted Alkenyl Sulfides via a Carbocupration of Alkynyl Sulfides with Aryl and Benzylic Diorganozincs

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Received 29 July 2011

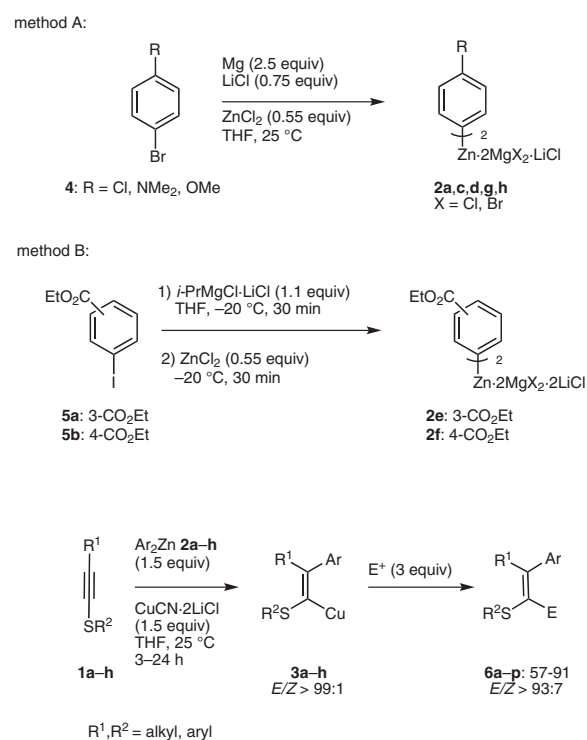
Abstract: A general method for the synthesis of tetrasubstituted alkenyl sulfides has been developed by the carbocupration of alkynyl sulfides with functionalized diorganozinc reagents in the presence of CuCN·2LiCl. The intermediate alkenylcopper reagents readily react with various electrophiles furnishing a broad range of highly functionalized alkenes with excellent stereoselectivity (*E/Z* 93:7–99:1).

Key words: alkynyl sulfides, copper, zinc, carbometalation, olefins

The stereoselective synthesis of tetrasubstituted olefins is of high interest, since these carbon fragments are present in many natural products and drugs like Tamoxifen.¹ The carbometalation of substituted alkynes constitutes an excellent stereoselective synthesis of such unsaturated molecules.² A variety of carbometalation reactions using copper, magnesium, boron, and tin reagents have been reported.³ Recently, the stereoselective carbometalation of alkynes using organozinc reagents and catalytic amounts of cobalt dibromide was also described.⁴ Organometallic zinc compounds are important reagents in organic synthesis, due to their high functional group compatibility.⁵ Furthermore, they can be transmetalated with copper(I) salts providing organocopper reagents that readily undergo carbometalation reactions.⁶

Herein, we wish to report a new approach to the stereoselective carbocupration of alkynes using highly functionalized aryl or benzylic diorganozinc reagents⁷ and stoichiometric amounts of copper(I) cyanide–di(lithium chloride) (CuCN·2LiCl).⁸ Various functional groups can be tolerated in the alkynyl side chain, such as cyanide, dimethylamino group, or bromide. Thus, the readily prepared alkynyl sulfides **1a–h** undergo smooth carbocupration with copper–zinc reagents Ar₂Cu(CN)ZnAr, prepared by mixing Ar₂Zn **2a–h** with CuCN·2LiCl (1.5 equiv) at 25 °C for 3–17 hours leading to the expected alkenylcopper compounds **3a–h** (Scheme 1) with excellent diastereoselectivity (*E/Z* > 99:1) as shown by GC-analysis of the crude hydrolyzed reaction mixture. The functionalized diarylzinc reagents of type **2** were prepared either by mag-

nesium insertion into aryl bromides of type **4** in the presence of zinc chloride (0.55 equiv; method A)^{7d,e} or by I/Mg-exchange using isopropylmagnesium chloride–lithium chloride⁹ as in the case of the aryl iodides **5a** and **5b** followed by transmetalation with zinc chloride (0.55 equiv; method B). The resulting alkenylcopper reagents **3a–h** reacted with various electrophiles affording alkenyl sulfides **6a–p** in 57–91% yield and *E/Z* ratios >93:7. Some loss of stereoselectivity was however observed during the quenching step.



Scheme 1 Preparation of diarylzinc reagents and their use in carbocupration reactions of alkynyl sulfides

Thus, the reaction of bis(4-chlorophenyl)zinc (**2a**) with the alkynyl sulfide **1a** in the presence of CuCN·2LiCl⁸ (1.5 equiv) provided the alkenylcopper intermediate **3a** at 25 °C within 16 hours. Its subsequent quenching with water led to the alkenyl sulfide **6a** in 80% yield; *E/Z* 97:3 (Table 1, entry 1). Similarly, the hydrolysis of the alkenylcopper compound **3b**, obtained after carbocupration with diphenylzinc (**2b**) and CuCN·2LiCl (1.5 equiv), afforded

the sulfide **6b** in 70% yield; *E/Z* 99:1 (entry 2).¹⁰ Iodolysis of the copper compounds **3a,c–e** led to the alkenyl iodides **6c–f** in 65–82% yield with excellent *E/Z* ratios > 99:1 (entries 3–6). The allylated sulfides **6g–j** were obtained by the reaction of the alkenylcopper intermediates **3e,f** with allyl bromide in 57–91% yield; *E/Z* 99:1 (entries 7–10). Furthermore, the alkynes **1d,g** underwent smooth carbocupration at 25 °C within eight hours with *meta*- or *para*-substituted organocopper reagents bearing an ester or a methoxy group, obtained after transmetalation of the diorganozinc reagents **2c,e,f** with CuCN·2LiCl (1.5 equiv). The resulting copper intermediates **3d,g** reacted

with allyl bromide or ethyl (2-bromomethyl)acrylate¹¹ furnishing the allylated alkenes **6k–m** in 65–87% yield; *E/Z* 93:7 (entries 11–13). Trapping the carbometalated alkenes **3d,h** with cyclohex-2-enone in the presence of trimethylsilyl chloride¹² afforded the Michael adducts **6n,o** in 60–80% yield with an excellent *E/Z* ratio of 99:1 (entries 14, 15). Finally, alkyne **1f** was carbocuprated with bis(3-fluorophenyl)zinc (**2h**) in the presence of CuCN·2LiCl within eight hours at 25 °C. Its subsequent acylation with 4-chlorobenzoyl chloride gave the desired ketone **6p** in 66% yield; *E/Z* 93:7 (entry 16).

Table 1 Carbocupration of Alkynes **1** and Reaction with Electrophiles

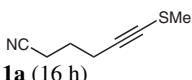
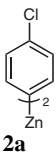
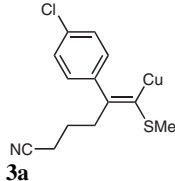
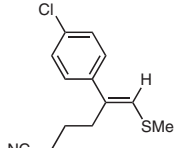
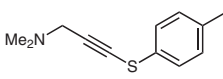
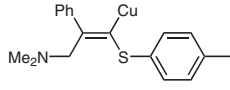
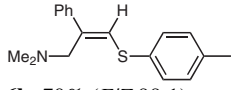
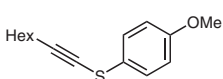
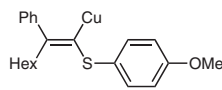
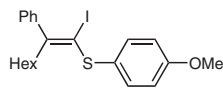
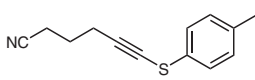
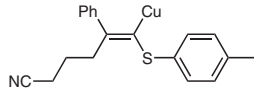
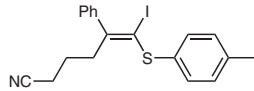
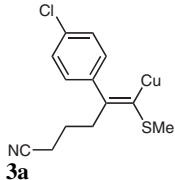
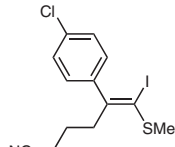
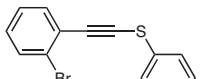
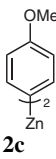
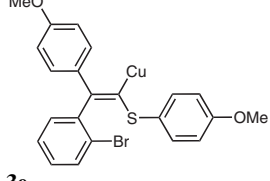
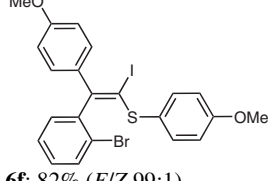
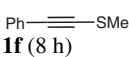
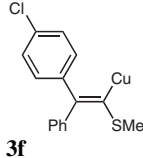
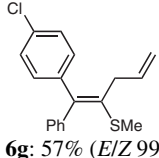
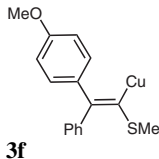
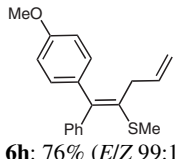
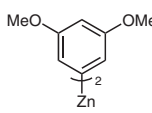
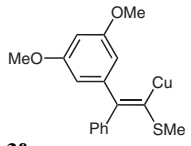
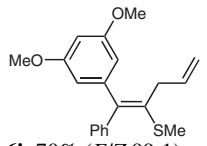
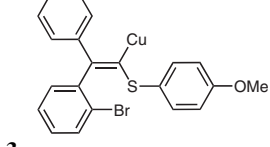
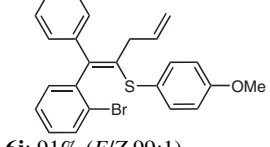
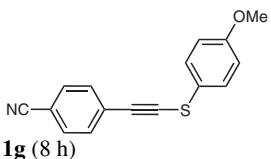
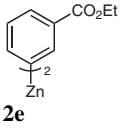
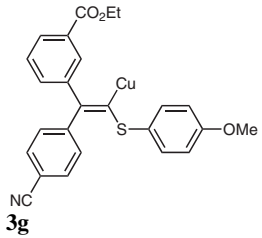
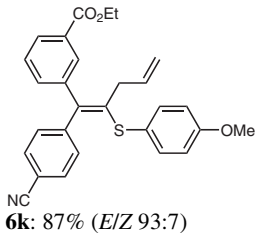
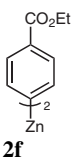
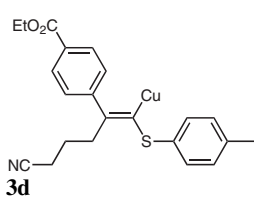
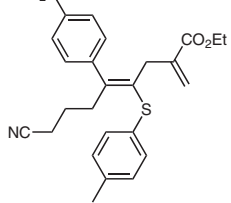
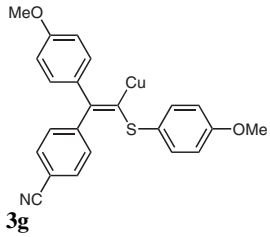
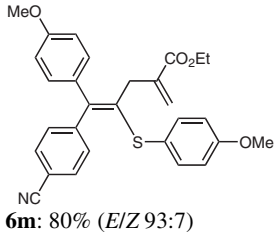
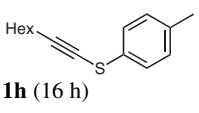
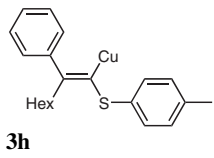
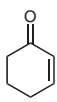
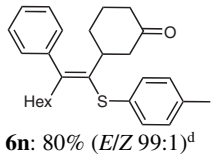
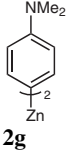
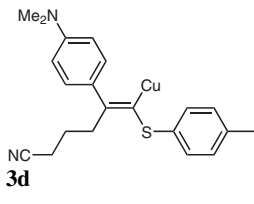
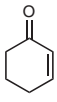
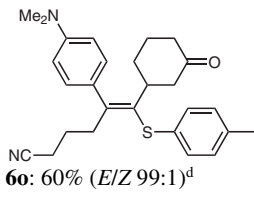
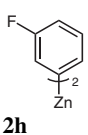
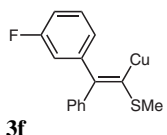
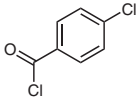
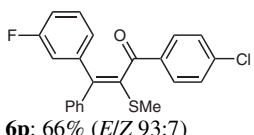
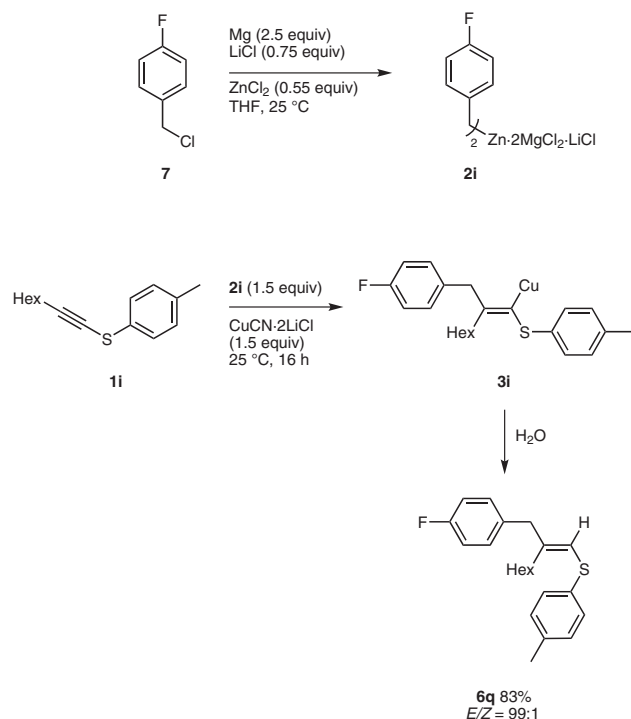
Entry	Alkyne 1 (time) ^a	Zinc reagent ^b	Copper intermediate 3	Electrophile	Product, yield (ratio) ^c
1	 1a (16 h)	 2a	 3a	H ₂ O	 6a : 80% (<i>E/Z</i> 97:3)
2	 1b (17 h)	Ph ₂ Zn 2b	 3b	H ₂ O	 6b : 70% (<i>E/Z</i> 99:1)
3	 1c (15 h)	2b	 3c	I ₂	 6c : 80% (<i>E/Z</i> 99:1)
4	 1d (8 h)	2b	 3d	I ₂	 6d : 65% (<i>E/Z</i> 99:1)
5	1a (16 h)	2a	 3a	I ₂	 6e : 70% (<i>E/Z</i> 99:1)
6	 1e (6 h)	 2c	 3e	I ₂	 6f : 82% (<i>E/Z</i> 99:1)
7	 1f (8 h)	2a	 3f		 6g : 57% (<i>E/Z</i> 99:1)
8	1f (8 h)	2c	 3f		 6h : 76% (<i>E/Z</i> 99:1)

Table 1 Carbocupration of Alkynes **1** and Reaction with Electrophiles (continued)

Entry	Alkyne 1 (time) ^a	Zinc reagent ^b	Copper intermediate 3	Electrophile	Product, yield (ratio) ^c
9	1f (8 h)		3f 		 6i : 70% (<i>E/Z</i> 99:1)
10	1e (6 h)	2c 	3e 		 6j : 91% (<i>E/Z</i> 99:1)
11	1g (8 h) 	2e 	3g 		 6k : 87% (<i>E/Z</i> 93:7)
12	1d (8 h)	2f 	3d 		 6l : 65% (<i>E/Z</i> 93:7)
13	1g (8 h)	2c 	3g 		 6m : 80% (<i>E/Z</i> 93:7)
14	1h (16 h) 	2b 	3h 		 6n : 80% (<i>E/Z</i> 99:1) ^d
15	1d (8 h)	2g 	3d 		 6o : 60% (<i>E/Z</i> 99:1) ^d
16	1f (8 h)	2h 	3f 		 6p : 66% (<i>E/Z</i> 93:7)

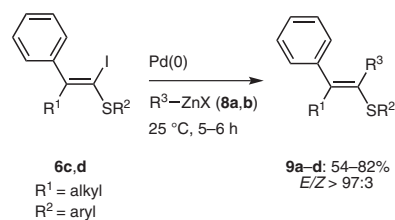
^a Reaction time for the carbometalation at 25 °C. First, the zinc reagent (1.5 equiv) was transmetalated using CuCN·2LiCl (1.5 equiv, –20 °C, 30 min).^b 2 MgX₂·2LiCl has been omitted for the sake of clarity; X = Cl, Br.^c Isolated yield of analytically pure product. *E/Z* ratio determined by 2D NMR.^d TMSCl (7.5 equiv) was used.

Besides arylzincs, it was also possible to use bis(benzyl)zinc reagents.⁴ Thus, bis(4-fluorobenzyl)zinc (**2i**, 1.5 equiv) was obtained by magnesium insertion to 4-fluorobenzyl chloride (**7**) in the presence of zinc chloride (0.55 equiv).¹³ The carbometalation reaction of the resulting diorganozinc reagent **2i** with the alkynyl sulfide **1i** in the presence of CuCN·2LiCl (1.5 equiv) provided the alkenylcopper intermediate **3i** at 25 °C in 16 hours. Its subsequent hydrolysis led to the benzylic sulfide **6q** in 83% yield; *E/Z* 99:1 (Scheme 2).



Scheme 2 Preparation of the dibenzylzinc reagent **2i** and carbocupration of alkynyl sulfide **1i**

The resulting alkenyl iodides **6c–d**, obtained by a carbocupration–iodolysis sequence, can be further used as electrophiles in palladium-catalyzed cross-coupling reactions (Scheme 3).¹⁴



Scheme 3 Palladium-catalyzed cross-coupling reaction of the alkenyl iodides **6c,d**

Negishi cross-coupling¹⁵ reaction of the alkenyl iodide **6c** using 2% palladium(II) acetate, 4% 2-(dicyclohexylphosphino)-2',6'-dimethoxybiphenyl (SPhos),¹⁶ and benzylzinc chloride (**8a**, 1.5 equiv) afforded the alkenyl sulfide **9a** in 77% yield; *E/Z* 98:2 (Table 2, entry 1). The sulfide

Table 2 Palladium-Catalyzed Cross-Coupling Reaction of Alkenyl Iodides **6c,d** with Zinc Reagents **8a,b** at 25 °C

Entry	Alkenyl iodide	Zinc reagent (time)	Product, yield (ratio) ^a
1	6c	8a (6 h) ^b	9a : 77% (<i>E/Z</i> 98:2)
2	6c	8b (5 h) ^c	9b : 54% (<i>E/Z</i> 97:3)
3	6d	8a (6 h) ^c	9c : 76% (<i>E/Z</i> 99:1)
4	6d	8b (5 h) ^c	9d : 82% (<i>E/Z</i> 99:1)

^a Isolated yield of analytically pure product. *E/Z* ratio determined by 2D NMR.

^b 2% Pd(OAc)₂ and 4% SPhos were used.

^c 3% Pd(dba)₂ and 6% P(*o*-furyl)₃ were used.

9b was obtained by using 3% bis(dibenzylideneacetone)palladium(0) [Pd(dba)₂], 6% tri(2-furyl)phosphine,¹⁷ and *p*-tolylzinc iodide (**8b**) within five hours at 25 °C in 54% yield; *E/Z* 97:3 (entry 2). The cross-coupling reaction of **6d** with benzylzinc chloride (**8a**) or *p*-tolylzinc iodide (**8b**) led exclusively to the alkenes **9c,d** in 76–82% yield (*E/Z* 99:1, entries 3 and 4).

In summary, we have reported a new copper(I)-mediated carbometalation reaction of alkenyl sulfides with functionalized aryl or benzylic diorganozinc reagents. Trapping reactions with various electrophiles yielded highly stereoselectively tetrasubstituted sulfides (*E/Z* 93:7 up to 99:1).

All reactions were carried out under an argon atmosphere in flame-dried glassware. Syringes which were used to transfer anhydrous solvents or reagents were purged with argon prior to use. THF was continuously refluxed and freshly distilled from Na benzophenone ketyl under N₂. Yields refer to isolated yields of compounds estimated to be >95% pure as determined by ¹H NMR (25 °C) and capillary GC. Purification via column chromatography was performed using silica gel 60 (40–63 μm 230–400 mesh ASTM); petroleum ether = PE. All reagents were obtained from commercial sources. Melting points were measured using a Büchi B-540 apparatus and are uncorrected. NMR spectra were recorded in CDCl₃, DMSO-*d*₆, acetone-*d*₆, or C₆D₆. Mass spectra and HRMS were recorded using electrospray ionization except where otherwise noted.

Alkynyl Sulfides **1**; General Procedure 1 (GP1)

A dry and argon-flushed Schlenk-flask, equipped with a magnetic stirring bar and septum, was charged with the alkyne dissolved in THF (1 mL per mmol) and cooled to –78 °C. Then, *n*-BuLi in hex-

ane (1.1 equiv) was added at -78°C and the mixture was stirred until deprotonation was complete. Then, the disulfide (1.2 equiv) was added at this temperature, and the resulting mixture was warmed to 25°C overnight. The mixture was quenched with sat. Na_2CO_3 (10 mL per mmol) and extracted with Et_2O (3×10 mL per mmol). The combined organic layers were dried (MgSO_4), filtered, and concentrated under reduced pressure. Purification by flash chromatography (silica gel) yielded the desired alkynyl sulfide **1**.

6-(Methylthio)hex-5-yne nitrile (**1a**)

Prepared according to GP1 from hexyne-5-nitrile (2.3 g, 25 mmol) in THF (13 mL) (metalation conditions: -78°C , 1.5 h). Purification by flash chromatography (pentane– Et_2O , 50:1) yielded **1a** (1 g, 30%) as a yellow oil.

IR (diamond-ATR, neat): 2928 (m), 2247 (m), 1735 (w), 1452 (m), 1424 (vs), 1312 (s), 978 (s), 960 (m), 765 cm^{-1} (w).

^1H NMR (300 MHz, CDCl_3): $\delta = 2.52\text{--}2.45$ (m, 4 H), $2.36\text{--}2.35$ (m, 3 H), $1.90\text{--}1.81$ (m, 2 H).

^{13}C NMR (75 MHz, CDCl_3): $\delta = 119.2$, 89.9 , 72.6 , 24.6 , 19.2 , 19.1 , 16.2 .

MS (EI, 70 eV): m/z (%) = 139 (M^+ , 65), 124 (21), 97 (28), 85 (100), 84 (33), 45 (14).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_7\text{H}_9\text{NS}$: 139.0456; found: 139.0450.

N,N-Dimethyl-3-[(4-methylphenyl)thio]prop-2-yn-1-amine (**1b**)

Prepared according to GP1 from *N,N*-dimethylprop-2-yn-1-amine (2.5 g, 30 mmol) in THF (15 mL) (metalation conditions: -78°C , 1 h). Purification by flash chromatography (pentane– Et_2O , 1:50 + 2 vol% Et_3N) yielded **1b** (4 g, 65%) as an orange oil.

IR (diamond-ATR, neat): 2972 (w), 1492 (m), 1451 (m), 1354 (m), 1322 (m), 1260 (w), 1084 (m), 1038 (s), 1016 (m), 1009 (m), 837 (m), 801 (vs), 670 cm^{-1} (w).

^1H NMR (400 MHz, $\text{DMSO}-d_6$): $\delta = 7.34\text{--}7.31$ (m, 2 H), $7.24\text{--}7.20$ (m, 2 H), 3.49 (s, 2 H), 2.27 (s, 3 H), 2.19 (m, 6 H).

^{13}C NMR (100 MHz, $\text{DMSO}-d_6$): $\delta = 136.3$, 130.1 , 128.4 , 125.9 , 95.3 , 70.0 , 48.4 , 43.6 , 20.5 .

MS (EI, 70 eV): m/z (%) = 205 (M^+ , 69), 204 (53), 172 (39), 162 (38), 161 (100), 147 (45), 117 (60), 114 (30), 91 (46).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{12}\text{H}_{15}\text{NS}$: 205.0925; found: 205.0922.

1-Methoxy-4-(oct-1-ynylthio)benzene (**1c**)

Prepared according to GP1 from oct-1-yne (2.2 g, 20 mmol) in THF (15 mL) (metalation conditions: -78°C , 2 h). Purification by flash chromatography (pentane + 2 vol% Et_3N) yielded **1c** (3 g, 60%) as a colorless oil.

IR (diamond-ATR, neat): 2929 (m), 1593 (w), 1492 (vs), 1462 (m), 1289 (m), 1243 (vs), 1174 (m), 1032 (s), 1006 (w), 820 (s), 623 cm^{-1} (w).

^1H NMR (300 MHz, CDCl_3): $\delta = 7.38\text{--}7.33$ (m, 2 H), $6.91\text{--}6.86$ (m, 2 H), 3.81 (s, 3 H), 2.41 (t, $J = 7.1$ Hz, 2 H), $1.63\text{--}1.30$ (m, 8 H), $0.93\text{--}0.88$ (m, 3 H).

^{13}C NMR (75 MHz, CDCl_3): $\delta = 158.7$, 128.3 , 124.0 , 114.9 , 98.4 , 66.0 , 55.4 , 31.3 , 28.6 , 22.5 , 20.3 , 14.0 .

MS (EI, 70 eV): m/z (%) = 248 (M^+ , 100), 140 (44), 139 (21), 133 (44).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{15}\text{H}_{20}\text{OS}$: 248.1234; found: 248.1231.

6-[(4-Methylphenyl)thio]hex-5-yne nitrile (**1d**)

Prepared according to GP1 from hex-5-yne nitrile (2.9 g, 30 mmol) in THF (15 mL) (metalation conditions: -78°C , 1.5 h). Purification by flash chromatography (pentane) yielded **1d** (3.2 g, 50%) as an orange oil.

IR (diamond-ATR, neat): 2942 (w), 2248 (w), 1491 (m), 1425 (m), 1210 (w), 1182 (w), 1117 (w), 1084 (m), 1016 (w), 803 cm^{-1} (vs).

^1H NMR (400 MHz, $\text{DMSO}-d_6$): $\delta = 7.34\text{--}7.30$ (m, 2 H), $7.23\text{--}7.19$ (m, 2 H), $2.60\text{--}2.56$ (m, 4 H), 2.28 (s, 3 H), $1.86\text{--}1.79$ (m, 2 H).

^{13}C NMR (100 MHz, $\text{DMSO}-d_6$): $\delta = 136.2$, 130.1 , 128.5 , 125.8 , 120.0 , 98.0 , 66.0 , 24.0 , 20.5 , 18.8 , 15.6 .

MS (EI, 70 eV): m/z (%) = 215 (M^+ , 100), 161 (41), 117 (44), 91 (14).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{13}\text{H}_{13}\text{NS}$: 215.0769; found: 215.0762.

1-Bromo-2-[(4-methoxyphenyl)thio]ethynyl]benzene (**1e**)

To 1-bromo-2-ethynylbenzene (958 mg, 5.29 mmol) in THF (5 mL) was added MeMgCl (1.77 mL, 5.29 mmol) at 25°C and the mixture was stirred for 1 h. Then, *S*-(4-methoxyphenyl)benzenesulfonothioate (1.48 g, 5.29 mmol) in THF (5 mL) was added at -40°C and the resulting mixture was slowly warmed to 0°C over 12 h. Purification according to GP1 yielded **1e** (1.48 g, 86%) as a yellow oil.

IR (diamond-ATR, neat): 2169 (w), 1590 (m), 1491 (s), 1290 (m), 1243 (vs), 1174 (s), 1025 (s), 820 (s), 749 (s), 681 cm^{-1} (m).

^1H NMR (300 MHz, CDCl_3): $\delta = 7.60\text{--}7.55$ (m, 1 H), $7.53\text{--}7.44$ (m, 3 H), $7.28\text{--}7.22$ (m, 1 H), $7.18\text{--}7.11$ (m, 1 H), $6.94\text{--}6.88$ (m, 2 H), 3.80 (s, 3 H).

^{13}C NMR (75 MHz, CDCl_3): $\delta = 159.1$, 133.0 , 132.4 , 129.2 , 129.0 , 127.0 , 125.3 , 125.0 , 122.5 , 115.1 , 95.0 , 82.4 , 55.4 .

MS (EI, 70 eV): m/z (%) = 318 (M^+ , 90), 305 (23), 239 (100), 195 (34), 152 (21), 140 (19), 43 (11).

HRMS: m/z [M] $^+$ calcd for $\text{C}_{15}\text{H}_{11}\text{BrOS}$: 317.9714; found: 317.9709.

[(Methylthio)ethynyl]benzene (**1f**)

Prepared according to the literature procedure.¹⁸

4-[(4-Methoxyphenyl)thio]ethynyl]benzonitrile (**1g**)

A dry and argon-flushed Schlenk-flask, equipped with a magnetic stirring bar and a septum, was charged with 4-ethynylbenzonitrile (1.3 g, 10 mmol) in THF (15 mL) and cooled to -78°C . Then, *i*-PrMgCl–LiCl (8.8 mL, 11 mmol) was slowly added at -78°C and the resulting mixture was stirred for 30 min. *S*-(4-Methoxyphenyl)benzenesulfonothioate (3.1 g, 11 mmol) in THF (10 mL) was added at this temperature, and the resulting mixture was allowed to warm up to 25°C overnight. Purification according to GP1 and flash chromatography (pentane + 2 vol% Et_3N) yielded **1g** (1.72 g, 65%) as a yellow solid; mp $88\text{--}90^{\circ}\text{C}$.

IR (diamond-ATR, neat): 2844 (w), 2224 (m), 2168 (m), 1590 (m), 1570 (m), 1492 (s), 1466 (m), 1458 (m), 1442 (m), 1408 (m), 1288 (m), 1244 (s), 1172 (s), 1120 (m), 1106 (m), 1022 (s), 850 (m), 828 (vs), 808 cm^{-1} (w).

^1H NMR (400 MHz, CDCl_3): $\delta = 7.59\text{--}7.57$ (m, 2 H), $7.50\text{--}7.47$ (m, 2 H), $7.44\text{--}7.40$ (m, 2 H), $6.94\text{--}6.90$ (m, 2 H), 3.81 (s, 3 H).

^{13}C NMR (100 MHz, CDCl_3): $\delta = 159.4$, 132.0 , 131.4 , 129.7 , 127.9 , 121.7 , 118.5 , 115.2 , 111.1 , 94.5 , 83.6 , 55.4 .

MS (EI, 70 eV): m/z (%) = 265 (M^+ , 100), 250 (48), 222 (14).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{16}\text{H}_{11}\text{NOS}$: 265.0561; found: 265.0556.

1-Methyl-4-(oct-1-ynylthio)benzene (1h)

Prepared according to GP1 from oct-1-yne (2.8 g, 25 mmol) in THF (25 mL) (metalation conditions: -78°C , 2 h). Purification by flash chromatography (pentane) yielded **1h** (3.2 g, 55%) as a colorless oil.

IR (diamond-ATR, neat): 2955 (m), 2927 (m), 2857 (m), 1492 (m), 1465 (w), 1456 (w), 1017 (w), 801 (vs), 725 cm^{-1} (w).

^1H NMR (400 MHz, $\text{DMSO}-d_6$): $\delta = 7.31\text{--}7.27$ (m, 2 H), $7.21\text{--}7.17$ (m, 2 H), 2.45 (t, $J = 7.0\text{ Hz}$, 2 H), 2.27 (s, 3 H), 1.54–1.47 (m, 2 H), 1.40–1.33 (m, 2 H), 1.29–1.22 (m, 4 H), 0.87–0.83 (m, 3 H).

^{13}C NMR (100 MHz, $\text{DMSO}-d_6$): $\delta = 136.0$, 130.0, 128.9, 125.5, 100.2, 64.4, 30.7, 28.0, 27.9, 22.0, 20.4, 19.5, 13.9.

MS (EI, 70 eV): m/z (%) = 232 (M^+ , 100), 163 (53), 161 (73), 130 (38), 124 (50), 117 (63), 91 (42), 67 (45).

HRMS (EI): m/z [$\text{M}]^+$ calcd for $\text{C}_{15}\text{H}_{20}\text{S}$: 232.1286; found: 232.1266.

Carbocupration of Alkynes with Functionalized Diorganozinc Reagents; General Procedure (GP2)

A dry and argon-flushed Schlenk-flask, equipped with a magnetic stirring bar and a septum, was charged with the diorganozinc reagent of type $\text{R}_2\text{Zn}\cdot 2\text{MgX}_2\cdot\text{LiCl}$ (1.5 equiv) and cooled to -20°C . $\text{CuCN}\cdot 2\text{LiCl}$ (1.5 equiv) was added dropwise and the resulting mixture was stirred for 30 min. Then, the alkynyl sulfide of type **1** was added, warmed to 25°C and stirred for the indicated time. The carbocupration progress was monitored by GC analysis of reaction aliquots, which were quenched with sat. NH_4Cl soln using dodecane as internal standard.

Purification of Tetrasubstituted Alkenes; General Procedure (P1)

The reaction mixture was quenched with sat. NH_4Cl –25% NH_3 in H_2O (4:1, 10 mL per mmol) and the resulting mixture was extracted with Et_2O ($3 \times 10\text{ mL}$ per mmol). The combined organic layers were dried (MgSO_4), filtered, and concentrated under reduced pressure.

(5E)-5-(4-Chlorophenyl)-6-(methylthio)hex-5-enenitrile (6a)

Prepared according to GP2 from **1a** (209 mg, 1.5 mmol) and bis(4-chlorophenyl)zinc (**2a**; 0.4 M in THF; 5.7 mL, 2.3 mmol) (carbometalation conditions: 25°C , 16 h). Purification according to P1 and flash chromatography (pentane– Et_2O , 9:1 + 2 vol% Et_3N) yielded **6a** (201 mg, 80%, E/Z 97:3) as a yellow oil.

IR (diamond-ATR, neat): 2924 (w), 1491 (m), 1424 (m), 1398 (w), 1091 (m), 1010 (m), 803 (vs), 646 cm^{-1} (w).

^1H NMR (400 MHz, CDCl_3): $\delta = 7.30\text{--}7.26$ (m, 2 H), $7.24\text{--}7.20$ (m, 2 H), 6.28 (s, 1 H), $2.72\text{--}2.69$ (m, 2 H), 2.38 (s, 3 H), 2.31 (t, $J = 7.2\text{ Hz}$, 2 H), 1.81–1.73 (m, 2 H).

^{13}C NMR (100 MHz, CDCl_3): $\delta = 138.7$, 134.5, 132.9, 128.8, 128.4, 126.8, 119.4, 30.3, 23.6, 17.4, 16.7.

MS (EI, 70 eV): m/z (%) = 251 (M^+ , 100), 199 (20), 197 (51), 195 (30), 163 (16), 149 (24), 115 (11).

HRMS (EI): m/z [$\text{M}]^+$ calcd for $\text{C}_{13}\text{H}_{14}\text{ClNS}$: 251.0535; found: 251.0534.

Dimethyl{(2Z)-3-[(4-methylphenyl)thio]-2-phenylprop-2-enyl}amine (6b)

Prepared according to GP2 from **1b** (205 mg, 1.0 mmol) and Ph_2Zn (**2b**, prepared by transmetalation of freshly titrated PhMgCl (1.43 M in THF; 4.2 mL, 6 mmol) with $\text{ZnCl}_2/\text{LiCl}$ (3 mL, 3 mmol) at 25°C for 10 min) (carbometalation conditions: 25°C , 17 h). Purification according to P1 and flash chromatographical purification (pentane– Et_2O , 5:1 + 2 vol% Et_3N) yielded **6b** (198 mg, 70%, E/Z 99:1) as a colorless oil.

IR (diamond-ATR, neat): 2970 (w), 2814 (m), 2764 (m), 1491 (s), 1454 (m), 1442 (m), 1176 (m), 1092 (m), 1041 (m), 1015 (m), 852 (m), 823 (m), 803 (s), 748 (vs), 694 cm^{-1} (s).

^1H NMR (400 MHz, $\text{DMSO}-d_6$): $\delta = 7.52\text{--}7.49$ (m, 2 H), $7.34\text{--}7.29$ (m, 4 H), $7.25\text{--}7.16$ (m, 3 H), 6.74 (s, 1 H), 3.44 (s, 2 H), 2.28 (s, 3 H), 2.15 (s, 6 H).

^{13}C NMR (100 MHz, $\text{DMSO}-d_6$): $\delta = 140.0$, 137.0, 136.3, 132.0, 130.0, 129.0, 128.2, 127.0, 126.0, 125.7, 58.7, 44.7, 20.6.

MS (EI, 70 eV): m/z (%) = 283 (M^+ , 54), 161 (13), 160 (100), 158 (15), 58 (80).

HRMS (EI): m/z [$\text{M}]^+$ calcd for $\text{C}_{18}\text{H}_{21}\text{NS}$: 283.1395; found: 283.1399.

1-[(1Z)-1-Iodo-2-phenyloct-1-enyl]thio]-4-methoxybenzene (6c); Typical Procedure (TP1)

Prepared according to GP2 from **1c** (994 mg, 4 mmol) and Ph_2Zn (**2b**, prepared by transmetalation of freshly titrated PhMgCl (1.43 M in THF; 8.4 mL, 12 mmol) with $\text{ZnCl}_2/\text{LiCl}$ (6 mL, 6 mmol) at 25°C for 10 min) (carbometalation conditions: 25°C , 15 h). Then, the mixture was cooled to -40°C and 1 M I_2 (5 g, 20 mmol) in THF soln was added. The soln was stirred for 10 min at this temperature. The reaction was quenched with sat. $\text{Na}_2\text{S}_2\text{O}_3$ (150 mL) and the resulting mixture was extracted with Et_2O ($3 \times 150\text{ mL}$). The combined organic layers were dried (MgSO_4), filtered, and concentrated under reduced pressure. Purification by flash chromatography (pentane + 2 vol% Et_3N) yielded **6c** (1.44 g, 80%, E/Z 99:1) as a yellow oil.

IR (diamond-ATR, neat): 2924 (m), 1592 (m), 1491 (s), 1452 (m), 1440 (m), 1286 (m), 1242 (s), 1179 (m), 1171 (m), 1031 (m), 825 (s), 773 (m), 746 (m), 698 (vs), 639 (m), 621 cm^{-1} (m).

^1H NMR (400 MHz, $\text{DMSO}-d_6$): $\delta = 7.40\text{--}7.36$ (m, 2 H), $7.33\text{--}7.28$ (m, 3 H), $7.18\text{--}7.16$ (m, 2 H), $7.02\text{--}6.98$ (m, 2 H), 3.77 (s, 3 H), 2.83 (d, $J = 7.2\text{ Hz}$, 2 H), 1.30–1.17 (m, 8 H), 0.81–0.77 (m, 3 H).

^{13}C NMR (100 MHz, $\text{DMSO}-d_6$): $\delta = 159.6$, 158.3, 146.0, 132.5, 128.7, 128.1, 128.0, 126.6, 115.5, 90.4, 55.7, 37.3, 31.4, 28.5, 27.9, 22.4, 14.3.

MS (EI, 70 eV): m/z (%) = 452 (M^+ , 16), 241 (31), 185 (100), 139 (82), 121 (19), 91 (17), 43 (12).

HRMS (EI): m/z [$\text{M}]^+$ calcd for $\text{C}_{21}\text{H}_{25}\text{IOS}$: 452.0671; found: 452.0673.

(5Z)-6-Iodo-6-[(4-methylphenyl)thio]-5-phenylhex-5-enenitrile (6d)

Prepared according to TP1 from **1d** (431 mg, 2 mmol) and Ph_2Zn (**2b**, prepared by transmetalation of freshly titrated PhMgCl (1.43 M in THF; 4.2 mL, 6 mmol) with $\text{ZnCl}_2/\text{LiCl}$ (3 mL, 3 mmol) at 25°C for 10 min) (carbometalation conditions: 25°C , 8 h). Purification by flash chromatography (pentane– Et_2O , 1:1 + 2 vol% Et_3N) yielded **6d** (273 mg, 65%, E/Z 99:1) as a white solid; mp $75\text{--}77^{\circ}\text{C}$.

IR (diamond-ATR, neat): 2927 (w), 1489 (m), 1453 (w), 1074 (w), 831 (m), 802 (s), 762 (m), 752 (m), 729 (w), 700 (vs), 675 cm^{-1} (w).

^1H NMR (400 MHz, $\text{DMSO}-d_6$): $\delta = 7.43\text{--}7.32$ (m, 3 H), $7.29\text{--}7.24$ (m, 6 H), 2.94–2.46 (m, 2 H), 2.52–2.46 (m, 2 H), 2.30 (s, 3 H), 1.59–1.51 (m, 2 H).

^{13}C NMR (100 MHz, $\text{DMSO}-d_6$): $\delta = 157.4$, 144.9, 136.9, 131.9, 130.0, 129.5, 128.3, 127.8, 127.7, 120.1, 89.2, 35.7, 23.5, 20.7, 15.7.

MS (EI, 70 eV): m/z (%) = 419 (M^+ , 12), 293 (20), 292 (100), 275 (27), 225 (27), 224 (67), 103 (10).

HRMS (EI): m/z [$\text{M}]^+$ calcd for $\text{C}_{19}\text{H}_{18}\text{INS}$: 419.0205; found: 419.0198.

(5Z)-5-(4-Chlorophenyl)-6-iodo-6-(methylthio)hex-5-enenitrile (6e)

Prepared according to TP1 from **1a** (209 mg, 1.5 mmol) and bis(4-chlorophenyl)zinc (**2a**; 0.4 M in THF; 5.7 mL, 2.3 mmol) (carbometalation conditions: 25 °C, 16 h). Purification by flash chromatography (pentane–Et₂O, 5:1 + 2 vol% Et₃N) yielded **6e** (264 mg, 70%, *E/Z* 99:1) as a yellow oil.

IR (diamond-ATR, neat): 2918 (w), 1482 (s), 1452 (m), 1421 (m), 1395 (m), 1167 (w), 1088 (s), 1013 (s), 969 (m), 835 (vs), 809 (s), 784 (m), 719 (m), 649 (m), 618 cm⁻¹ (m).

¹H NMR (400 MHz, CDCl₃): δ = 7.35 (d, *J* = 8.2 Hz, 2 H), 7.05 (d, *J* = 8.2 Hz, 2 H), 2.92–2.88 (m, 2 H), 2.35 (s, 3 H), 2.31 (t, *J* = 7.2 Hz, 2 H), 1.72–1.64 (m, 2 H).

¹³C NMR (100 MHz, CDCl₃): δ = 151.2, 143.8, 133.8, 129.4, 128.8, 119.0, 96.9, 36.1, 23.7, 22.7, 16.6.

MS (EI, 70 eV): *m/z* (%) = 377 (M⁺, 12), 252 (39), 251 (19), 250 (100), 175 (26), 167 (21), 137 (12).

HRMS (EI): *m/z* [M]⁺ calcd for C₁₃H₁₃ClINS: 376.9502; found: 376.9490.

1-Bromo-2-[(E)-2-iodo-1-(4-methoxyphenyl)-2-[(4-methoxyphenyl)thio]vinyl]benzene (6f)

Prepared according to TP1 from **1e** (638 mg, 2 mmol) and bis(4-methoxyphenyl)zinc (**2c**; 0.37 M in THF; 8.11 mL, 3 mmol) (carbometalation conditions: 25 °C, 6 h). Purification by flash chromatography (pentane–Et₂O, 3:1 + 2 vol% Et₃N) yielded **6f** (908 mg, 82%, *E/Z* 99:1) as a yellow oil.

IR (diamond-ATR, neat): 2956 (w), 1590 (m), 1504 (m), 1492 (s), 1462 (m), 1440 (m), 1288 (m), 1244 (vs), 1172 (s), 1028 (s), 826 (s), 770 (m), 742 (s), 708 (w), 640 cm⁻¹ (w).

¹H NMR (400 MHz, acetone-*d*₆): δ = 7.63 (dd, *J* = 8.1 Hz, 1.1 Hz, 1 H), 7.51 (dd, *J* = 7.6, 1.8 Hz, 1 H), 7.45–7.37 (m, 5 H), 7.27–7.21 (m, 1 H), 6.99–6.93 (m, 2 H), 6.91–6.86 (m, 2 H), 3.82 (s, 3 H), 3.78 (s, 3 H).

¹³C NMR (100 MHz, acetone-*d*₆): δ = 161.1, 160.3, 152.4, 143.2, 136.7, 135.1, 133.7, 131.4, 131.0, 130.1, 128.6, 127.6, 122.9, 115.5, 114.0, 97.7, 55.7, 55.5.

MS (EI, 70 eV): *m/z* (%) = 552 (M⁺, 14), 473 (24), 427 (69), 346 (100), 331 (24), 139 (24).

HRMS (EI): *m/z* [M]⁺ calcd for C₂₂H₁₈BrIO₂S: 551.9256; found: 551.9249.

1-Chloro-4-[(1Z)-2-(methylthio)-1-phenylpenta-1,4-dienyl]benzene (6g); Typical Procedure (TP2)

Prepared according to GP2 from **1f** (373 mg, 2 mmol) and bis(4-chlorophenyl)zinc (**2a**; 0.55 M in THF; 5.4 mL, 3 mmol) (carbometalation conditions: 25 °C, 3 h). Then, the mixture was cooled to –40 °C and allyl bromide (0.58 mL, 6 mmol) was added. The soln was stirred for 30 min at this temperature followed by 30 min at 0 °C. Purification according to P1 and flash chromatographical purification (pentane + 2 vol% Et₃N) yielded **6g** (343 mg, 57%, *E/Z* 99:1) as a white solid; mp 55–56 °C.

IR (diamond-ATR, neat): 2982 (w), 1485 (s), 1440 (m), 1426 (m), 1391 (m), 1087 (s), 1070 (m), 1013 (s), 989 (m), 915 (vs), 836 (vs), 825 (m), 779 (s), 759 (vs), 721 (s), 701 (vs), 618 (m), 608 cm⁻¹ (m).

¹H NMR (400 MHz, C₆D₆): δ = 7.32–7.29 (m, 2 H), 7.18–7.13 (m, 2 H), 7.06–7.00 (m, 3 H), 6.94–6.91 (m, 2 H), 5.90–5.80 (m, 1 H), 5.18–5.12 (m, 1 H), 5.08–5.04 (m, 1 H), 2.97–2.95 (m, 2 H), 1.78 (s, 3 H).

¹³C NMR (100 MHz, C₆D₆): δ = 142.5, 141.3, 139.0, 136.2, 134.7, 133.0, 130.9, 130.2, 128.6, 128.4, 127.2, 116.2, 36.9, 15.0.

MS (EI, 70 eV): *m/z* (%) = 300 (M⁺, 100), 253 (20), 252 (20), 246 (18), 244 (59), 218 (20), 217 (26), 125 (16), 91 (14).

HRMS (EI): *m/z* [M]⁺ calcd for C₁₈H₁₇ClIS: 300.0739; found: 300.0740.

1-Methoxy-4-[(1Z)-2-(methylthio)-1-phenylpenta-1,4-dienyl]benzene (6h)

Prepared according to TP2 from **1f** (373 mg, 2 mmol) and bis(4-methoxyphenyl)zinc (**2c**; 0.37 M in THF; 8.11 mL, 3 mmol) (carbometalation conditions: 25 °C, 8 h). Purification by flash chromatography (pentane + 2 vol% Et₃N) yielded **6h** (451 mg, 76%, *E/Z* 99:1) as a yellow oil.

IR (diamond-ATR, neat): 2969 (w), 1739 (s), 1604 (m), 1506 (vs), 1440 (m), 1365 (m), 1281 (m), 1242 (vs), 1217 (s), 1172 (m), 1031 (m), 832 (s), 799 (m), 774 (m), 740 (m), 697 cm⁻¹ (vs).

¹H NMR (300 MHz, C₆D₆): δ = 7.45–7.41 (m, 2 H), 7.21–7.12 (m, 4 H), 7.08–7.02 (m, 1 H), 6.72–6.67 (m, 2 H), 6.05–5.92 (m, 1 H), 5.27–5.20 (m, 1 H), 5.13–5.08 (m, 1 H), 3.27 (s, 3 H), 3.17–3.14 (m, 2 H), 1.84 (s, 3 H).

¹³C NMR (75 MHz, C₆D₆): δ = 159.0, 143.4, 140.7, 136.7, 135.4, 133.2, 130.7, 130.3, 128.3, 127.0, 116.0, 113.9, 54.7, 37.3, 15.2.

MS (EI, 70 eV): *m/z* (%) = 297 (17), 296 (M⁺, 100), 249 (13), 248 (10), 240 (53), 225 (13).

HRMS (EI): *m/z* [M]⁺ calcd for C₁₉H₂₀OS: 296.1235; found: 296.1227.

1,3-Dimethoxy-5-[(1E)-2-(methylthio)-1-phenylpenta-1,4-dienyl]benzene (6i)

Prepared according to TP2 from **1f** (373 mg, 2 mmol) and bis(3,5-dimethoxyphenyl)zinc (**2d**; 0.54 M in THF; 5.5 mL, 3 mmol) (carbometalation conditions: 25 °C, 7 h). Purification by flash chromatography (pentane + 2 vol% Et₃N) yielded **6i** (457 mg, 70%, *E/Z* 99:1) as a yellow oil.

IR (diamond-ATR, neat): 1581 (s), 1449 (m), 1418 (m), 1342 (m), 1264 (m), 1201 (s), 1148 (vs), 1064 (m), 1047 (m), 908 (m), 849 (m), 836 (s), 813 (m), 706 (vs), 694 (s), 664 cm⁻¹ (m).

¹H NMR (400 MHz, C₆D₆): δ = 7.48–7.46 (m, 2 H), 7.14–7.12 (m, 2 H), 7.03–6.99 (m, 1 H), 6.58 (d, *J* = 2.3 Hz, 2 H), 6.45 (t, *J* = 2.2 Hz, 1 H), 5.99–5.89 (m, 1 H), 5.23–5.18 (m, 1 H), 5.07–5.03 (m, 1 H), 3.24 (s, 6 H), 3.16–3.14 (m, 2 H), 1.79 (s, 3 H).

¹³C NMR (100 MHz, C₆D₆): δ = 161.3, 145.0, 142.6, 140.6, 136.7, 134.0, 130.0, 128.4, 127.1, 116.0, 107.7, 99.6, 54.8, 37.3, 15.0.

MS (EI, 70 eV): *m/z* (%) = 326 (M⁺, 100), 311 (25), 279 (44), 278 (86), 270 (42), 264 (25), 203 (14), 139 (13), 91 (11).

HRMS (EI): *m/z* [M]⁺ calcd for C₂₀H₂₂O₂S: 326.1341; found: 326.1338.

1-Bromo-2-[(1Z)-1-(4-methoxyphenyl)-2-[(4-methoxyphenyl)thio]penta-1,4-dienyl]benzene (6j)

Prepared according to TP2 from **1e** (318 mg, 1 mmol) and bis(4-methoxyphenyl)zinc (**2c**; 0.37 M in THF; 4.05 mL, 1.5 mmol) (carbometalation conditions: 25 °C, 6 h). Purification by flash chromatography (pentane–Et₂O, 12:1 + 2 vol% Et₃N) yielded **6j** (426 mg, 91%, *E/Z* 99:1) as a yellow oil.

IR (diamond-ATR, neat): 1592 (m), 1508 (m), 1492 (s), 1462 (m), 1440 (m), 1284 (m), 1242 (vs), 1172 (s), 1104 (m), 1028 (s), 914 (m), 828 (s), 800 (m), 766 (m), 742 (s), 626 cm⁻¹ (w).

¹H NMR (400 MHz, acetone-*d*₆): δ = 7.63–7.59 (m, 1 H), 7.47–7.43 (m, 1 H), 7.42–7.36 (m, 3 H), 7.34–7.28 (m, 2 H), 7.22–7.16 (m, 1 H), 6.93–6.84 (m, 4 H), 5.89–5.78 (m, 1 H), 5.06–4.97 (m, 2 H), 3.79 (s, 3 H), 3.76 (s, 3 H), 3.01–2.96 (m, 2 H).

^{13}C NMR (100 MHz, acetone- d_6): δ = 160.7, 159.8, 144.7, 141.6, 136.5, 136.0, 135.7, 133.6, 133.0, 132.1, 131.1, 129.5, 128.4, 124.6, 124.1, 116.6, 115.4, 114.2, 55.6, 55.5, 36.1.

MS (EI, 70 eV): m/z (%) = 466 (M^+ , 6), 387 (100), 346 (39), 279 (1), 215 (2), 139 (2).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{25}\text{H}_{23}\text{BrO}_2\text{S}$: 466.0602; found: 466.0600.

Ethyl 3-[(1Z)-1-(4-Cyanophenyl)-2-[(4-methoxyphenyl)thio]penta-1,4-dienyl]benzoate (6k)

Prepared according to TP2 from **1g** (265 mg, 1 mmol) and bis[3-(ethoxycarbonyl)phenyl]zinc (**2e**, prepared by adding *i*-PrMgCl-LiCl (3 mmol) to 1.0 M ethyl 3-iodobenzoate (828 mg, 3 mmol) in THF at -20°C and stirring for 0.5 h and further transmetalation with ZnCl_2 (1.5 mL, 1.5 mmol) at -20°C for 10 min) (carbometalation conditions: 25°C , 8 h). Purification by flash chromatography (pentane-Et $_2$ O, 19:1 + 2 vol% Et $_3$ N) yielded **6k** (395 mg, 87%, *E/Z* 93:7) as a yellow oil.

IR (diamond-ATR, neat): 1714 (s), 1592 (m), 1492 (s), 1462 (m), 1284 (s), 1244 (vs), 1222 (m), 1172 (s), 1104 (m), 1082 (m), 1026 (m), 914 (m), 826 (s), 764 (m), 740 (m), 708 (m), 628 cm^{-1} (w).

^1H NMR (400 MHz, DMSO- d_6): δ = 7.88–7.85 (m, 1 H), 7.82–7.78 (m, 3 H), 7.53–7.48 (m, 4 H), 7.37–7.33 (m, 2 H), 6.95–6.91 (m, 2 H), 5.80–5.70 (m, 1 H), 5.05–5.02 (m, 1 H), 4.92–4.87 (m, 1 H), 4.28 (q, J = 7.1 Hz, 2 H), 3.74 (s, 3 H), 2.80 (d, J = 6.2 Hz, 2 H), 1.28 (t, J = 7.2 Hz, 3 H).

^{13}C NMR (100 MHz, DMSO- d_6): δ = 165.3, 159.3, 146.5, 140.9, 140.0, 135.7, 134.7, 134.4, 133.5, 132.2, 130.3, 130.1, 129.2, 129.0, 128.2, 122.5, 118.7, 116.7, 114.9, 109.8, 60.8, 55.2, 35.9, 14.1.

MS (EI, 70 eV): m/z (%) = 456 (36), 455 (M^+ , 100), 140 (21), 139 (10).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{28}\text{H}_{27}\text{NO}_4\text{S}$: 455.1555; found: 455.1563.

Ethyl 4-[(1E)-1-(3-Cyanopropyl)-4-(ethoxycarbonyl)-2-[(4-methylphenyl)thio]penta-1,4-dienyl]benzoate (6l); Typical Procedure (TP3)

Prepared according to GP2 from **1d** (215 mg, 1 mmol) and bis[4-(ethoxycarbonyl)phenyl]zinc (**2f**, prepared by adding *i*-PrMgCl-LiCl (1.96 mL, 2.95 mmol) to 1 M ethyl 4-iodobenzoate (828 mg, 3 mmol) in THF at -20°C and stirring for 0.5 h and further transmetalation with ZnCl_2 (1.5 mL, 1.5 mmol) at -20°C for 10 min) (carbometalation conditions: 25°C , 8 h). Then, the mixture was cooled to -40°C and ethyl 2-(bromomethyl)acrylate¹¹ (580 mg, 3 mmol) was added. Purification according to P1 and flash chromatographical purification (pentane-Et $_2$ O, 1:1 + 2 vol% Et $_3$ N) yielded **6l** (273 mg, 65%, *E/Z* 93:7) as a yellow oil.

IR (diamond-ATR, neat): 1711 (vs), 1403 (m), 1367 (m), 1271 (vs), 1221 (m), 1176 (m), 1138 (s), 1102 (s), 1018 (s), 947 (m), 862 (m), 808 (s), 780 (m), 712 cm^{-1} (m).

^1H NMR (400 MHz, DMSO- d_6): δ = 7.96–7.93 (m, 2 H), 7.41–7.38 (m, 2 H), 7.23–7.15 (m, 4 H), 6.09 (d, J = 0.9 Hz, 1 H), 5.51 (d, J = 1.2 Hz, 1 H), 4.30 (q, J = 7.1 Hz, 2 H), 3.94 (q, J = 7.0 Hz, 2 H), 2.89–2.85 (m, 4 H), 2.48–2.45 (m, 2 H), 2.28 (s, 3 H), 1.57–1.50 (m, 2 H), 1.30 (t, J = 7.0 Hz, 3 H), 1.09 (t, J = 7.0 Hz, 3 H).

^{13}C NMR (100 MHz, DMSO- d_6): δ = 165.6, 165.3, 145.6, 145.5, 136.8, 136.6, 130.5, 129.9, 129.8, 129.2, 128.9, 128.9, 128.1, 126.1, 120.2, 60.7, 60.3, 35.0, 34.4, 23.5, 20.6, 15.9, 14.1, 13.8.

MS (EI, 70 eV): m/z (%) = 477 (M^+ , 20), 365 (25), 354 (51), 308 (100), 284 (35), 280 (55), 279 (33), 238 (72), 129 (15), 124 (19), 86 (18).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{28}\text{H}_{31}\text{NO}_4\text{S}$: 477.1974; found: 477.1970.

Ethyl (4E)-5-(4-Cyanophenyl)-5-(4-methoxyphenyl)-4-[(4-methoxyphenyl)thio]-2-methylenepent-4-enoate (6m)

Prepared according to TP3 from **1g** (265 mg, 1 mmol) and bis(4-methoxyphenyl)zinc (**2c**; 0.37 M in THF; 4 mL, 1.5 mmol) (carbometalation conditions: 25°C , 8 h). Purification by flash chromatography (pentane-Et $_2$ O, 1:1 + 2 vol% Et $_3$ N) yielded **6m** (388 mg, 80%, *E/Z* 93:7) as a yellow oil.

IR (diamond-ATR, neat): 2836 (w), 1712 (s), 1604 (m), 1592 (m), 1508 (m), 1492 (s), 1462 (m), 1284 (m), 1242 (vs), 1172 (s), 1136 (s), 1106 (m), 1028 (s), 828 (s), 808 (m), 618 cm^{-1} (w).

^1H NMR (400 MHz, DMSO- d_6): δ = 7.81–7.78 (m, 2 H), 7.49–7.48 (m, 2 H), 7.26–7.11 (m, 2 H), 6.91–6.87 (m, 4 H), 6.20 (d, J = 0.9 Hz, 1 H), 5.71 (d, J = 0.9 Hz, 1 H), 3.98 (q, J = 7.2 Hz, 2 H), 3.73 (s, 3 H), 3.71 (s, 3 H), 3.09 (s, 2 H), 1.09 (t, J = 7.2 Hz, 3 H).

^{13}C NMR (100 MHz, DMSO- d_6): δ = 165.5, 159.2, 158.5, 147.0, 142.2, 136.6, 134.4, 132.6, 132.6, 132.1, 130.1, 129.6, 126.3, 122.6, 118.7, 114.7, 113.9, 109.6, 60.3, 55.2, 55.0, 33.6, 13.9.

MS (EI, 70 eV): m/z (%) = 485 (M^+ , 100), 346 (63), 345 (50), 317 (32), 316 (75), 272 (23), 140 (28).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{29}\text{H}_{27}\text{NO}_4\text{S}$: 485.1661; found: 485.1651.

3-[(1E)-1-[(4-Methylphenyl)thio]-2-phenyloct-1-enyl]cyclohexanone (6n); Typical Procedure (TP4)

Prepared according to GP2 from **1h** (465 mg, 2 mmol) and Ph_2Zn [**2b**, prepared by transmetalation of freshly titrated PhMgCl (1.43 M in THF; 4.2 mL, 6 mmol) with $\text{ZnCl}_2/\text{LiCl}$ (3 mL, 3 mmol) at 25°C for 10 min] (carbometalation conditions: 25°C , 10 h). Then, the mixture was cooled to -40°C and a premixed soln of TMSCl (2.24 mL, 17.5 mmol) and cyclohex-2-enone (0.68 mL, 7 mmol) was added. The soln was stirred for at this temperature and slowly warmed to 0°C overnight. Purification according to P1 and flash chromatographical purification (pentane-Et $_2$ O, 5:1 + 2 vol% Et $_3$ N) yielded **6n** (651 mg, 80%, *E/Z* 99:1) as a colorless oil.

IR (diamond-ATR, neat): 2953 (m), 2926 (m), 2856 (m), 1712 (vs), 1490 (s), 1446 (m), 1258 (m), 1222 (m), 1117 (m), 1080 (m), 803 (s), 777 (m), 702 cm^{-1} (vs).

^1H NMR (400 MHz, DMSO- d_6): δ = 7.43–7.39 (m, 2 H), 7.34–7.27 (m, 1 H), 7.21–7.18 (m, 4 H), 7.15–7.13 (m, 2 H), 2.73–2.65 (m, 1 H), 2.61–2.51 (m, 3 H), 2.25 (s, 3 H), 2.22–2.17 (m, 1 H), 1.99–1.91 (m, 2 H), 1.85–1.79 (m, 1 H), 1.73–1.66 (m, 1 H), 1.55–1.52 (m, 1 H), 1.24–1.15 (m, 1 H), 1.10–1.00 (m, 8 H), 0.74–0.71 (m, 3 H).

^{13}C NMR (100 MHz, DMSO- d_6): δ = 209.7, 153.0, 144.8, 140.9, 134.4, 133.9, 130.7, 129.8, 128.5, 127.3, 127.3, 126.0, 45.8, 44.1, 37.0, 30.9, 29.5, 28.4, 26.7, 24.3, 21.8, 20.4, 13.8.

MS (EI, 70 eV): m/z (%) = 407 (27), 406 (M^+ , 100), 335 (19), 283 (14), 91 (12).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{27}\text{H}_{34}\text{OS}$: 406.2330; found: 406.2321.

(5E)-5-[4-(Dimethylamino)phenyl]-6-[(4-methylphenyl)thio]-6-(3-oxocyclohexyl)hex-5-enenitrile (6o)

Prepared according to TP4 from **1d** (217 mg, 1 mmol) and bis[4-(dimethylamino)phenyl]zinc (**2g**; 0.42 M in THF; 3.6 mL, 1.5 mmol) (carbometalation conditions: 25°C , 8 h). Purification by flash chromatography (pentane-Et $_2$ O, 1:1 + 2 vol% Et $_3$ N) yielded **6o** (260 mg, 60%, *E/Z* 99:1) as a yellow oil.

IR (diamond-ATR, neat): 2932 (m), 2860 (m), 1708 (vs), 1608 (vs), 1518 (vs), 1490 (s), 1446 (m), 1350 (s), 1258 (m), 1222 (s), 1196 (m), 1116 (m), 1062 (m), 946 (m), 820 (s), 804 (vs), 736 cm^{-1} (m).

^1H NMR (400 MHz, DMSO- d_6): δ = 7.18–7.13 (m, 4 H), 7.07–7.04 (m, 2 H), 6.76–6.72 (m, 2 H), 2.91 (s, 6 H), 2.72–2.62 (m, 2 H), 2.57–2.51 (m, 1 H), 2.29 (t, J = 7.1 Hz, 2 H), 2.26 (s, 3 H), 2.23–

2.18 (m, 1 H), 2.02–1.92 (m, 2 H), 1.88–1.82 (m, 1 H), 1.77–1.67 (m, 1 H), 1.58–1.54 (m, 1 H), 1.35–1.22 (m, 4 H).

^{13}C NMR (100 MHz, $\text{DMSO}-d_6$): δ = 209.9, 151.9, 149.5, 134.4, 134.0, 130.9, 129.9, 128.2, 127.1, 125.9, 120.3, 112.0, 45.9, 45.7, 44.1, 36.1, 29.5, 24.3, 23.3, 20.5, 15.8, 11.6.

MS (EI, 70 eV): m/z (%) = 433 (29), 432 (M^+ , 100), 309 (17), 251 (10).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{27}\text{H}_{32}\text{N}_2\text{OS}$: 432., 2235; found: 432.2232.

(2Z)-1-(4-Chlorophenyl)-3-(3-fluorophenyl)-2-(methylthio)-3-phenylprop-2-en-1-one (6p)

Prepared according to GP2 from **1f** (373 mg, 2 mmol) and bis(3-fluorophenyl)zinc (**2h**; 0.41 M in THF; 7.3 mL, 3 mmol) (carbometalation conditions: 25 °C, 8 h). Then, the mixture was cooled to –78 °C and 4-chlorobenzoyl chloride (0.77 mL, 6 mmol) was added. The soln was slowly warmed to –20 °C overnight. Purification according to P1 and flash chromatographical purification (pentane– Et_2O , 98:2 + 2 vol% Et_3N) yielded **6p** (505 mg, 66%, *E/Z* 93:7) as a white solid; mp 79–82 °C.

IR (diamond-ATR, neat): 1660 (s), 1581 (s), 1569 (m), 1479 (m), 1426 (m), 1398 (m), 1233 (m), 1214 (s), 1167 (m), 1093 (s), 1084 (m), 1010 (m), 914 (m), 837 (m), 789 (s), 752 (m), 722 (s), 696 (vs), 680 (m), 628 (s), 610 cm^{-1} (m).

^1H NMR (300 MHz, C_6D_6): δ = 7.83–7.79 (m, 2 H), 7.37–7.34 (m, 2 H), 7.20–7.18 (m, 2 H), 7.12–7.08 (m, 1 H), 7.02–6.98 (m, 1 H), 6.94–6.87 (m, 3 H), 6.61–6.55 (m, 1 H), 6.45–6.40 (m, 1 H), 1.80 (s, 3 H).

^{13}C NMR (75 MHz, C_6D_6): δ = 191.7, 162.6 (d, J = 246.9 Hz), 142.9 (d, J = 7.5 Hz), 140.4 (d, J = 2.1 Hz), 140.0, 139.9, 134.8, 134.4, 130.9, 130.2, 129.8 (d, J = 8.3 Hz), 129.1, 128.6, 128.4, 125.9 (d, J = 2.8 Hz), 117.0 (d, J = 21.9 Hz), 114.9 (d, J = 21.1 Hz), 15.9.

MS (EI, 70 eV): m/z (%) = 382 (M^+ , 100), 243 (57), 228 (69), 197 (24), 196 (28), 139 (62), 111 (23).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{22}\text{H}_{16}\text{ClFOS}$: 382.0594; found: 382.0590.

1-Fluoro-4-{2-hexyl-3-[(4-methylphenyl)thio]prop-2-enyl}benzene (6q)

Prepared according to GP2 from **1i** (232 mg, 1 mmol) and bis(4-fluorobenzyl)zinc (**2i**; 0.38 M in THF; 3.9 mL, 1.5 mmol) (carbometalation conditions: 25 °C, 16 h). Purification according to P1 and flash chromatographical purification (pentane– Et_2O , 9:1 + 2 vol% Et_3N) yielded **6q** (284 mg, 83%, *E/Z* 99:1) as a colorless oil.

IR (diamond-ATR, neat): 2954 (w), 2926 (m), 2856 (w), 1508 (vs), 1492 (m), 1466 (w), 1220 (vs), 1156 (m), 1092 (m), 1016 (w), 820 (s), 804 (s), 734 cm^{-1} (w).

^1H NMR (400 MHz, CDCl_3): δ = 7.20–7.08 (m, 6 H), 7.00–6.94 (m, 2 H), 5.89 (s, 1 H), 3.41 (s, 2 H), 2.31 (s, 3 H), 2.20–2.16 (m, 2 H), 1.42–1.23 (m, 8 H), 0.89–0.85 (m, 3 H).

^{13}C NMR (100 MHz, CDCl_3): δ = 161.5 (d, J = 244.2 Hz), 143.9 (d, J = 0.8 Hz), 135.9, 134.8 (d, J = 3.1 Hz), 133.3, 130.4 (d, J = 8.1 Hz), 129.7, 128.7, 119.5, 115.2 (d, J = 21.5 Hz), 42.4, 31.7, 31.6, 29.2, 27.7, 22.6, 20.9, 14.1.

MS (EI, 70 eV): m/z (%) = 343 (23), 342 (M^+ , 100), 147 (17), 135 (13), 109 (31).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{22}\text{H}_{27}\text{FS}$: 342.1817; found: 342.1809.

1-[(1E)-1-Benzyl-2-phenyloct-1-enyl]thio]-4-methoxybenzene (9a)

In a dry and argon-flushed Schlenk flask equipped with a septum and a magnetic stirring bar, **6c** (452 mg, 1 mmol) was dissolved in THF (1 mL). Then, $\text{Pd}(\text{OAc})_2$ (4.5 mg, 2 mol%), SPhos^{16} (16.4 mg, 4 mol%), and BnZnCl (**8a**; 1.19 M in THF; 1.36 mL, 1.5 mmol) were added at 0 °C and the mixture was warmed to 25 °C and stirred at this temperature for 6 h. The mixture was quenched with sat. NH_4Cl soln (10 mL) and the resulting mixture was extracted with Et_2O (3×10 mL). The combined organic layers were dried (MgSO_4), filtered, and concentrated under reduced pressure. Purification by flash chromatography (pentane + 2 vol% Et_3N) yielded **9a** (112 mg, 77%, *E/Z* 98:2) as a colorless oil.

IR (diamond-ATR, neat): 2925 (m), 1592 (m), 1492 (s), 1462 (m), 1452 (m), 1440 (m), 1286 (m), 1243 (vs), 1179 (m), 1171 (m), 1031 (m), 825 (s), 747 (m), 698 cm^{-1} (vs).

^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ = 7.37–7.32 (m, 3 H), 7.29–7.25 (m, 2 H), 7.23–7.11 (m, 7 H), 6.91–6.88 (m, 2 H), 3.73 (s, 3 H), 3.26 (s, 2 H), 2.80–2.76 (m, 2 H), 1.28–1.17 (m, 8 H), 0.81–0.78 (m, 3 H).

^{13}C NMR (100 MHz, $\text{DMSO}-d_6$): δ = 159.0, 147.5, 141.9, 139.6, 132.8, 129.7, 128.8, 128.6, 128.5, 128.3, 127.5, 126.3, 124.9, 115.3, 55.6, 46.1, 36.6, 31.4, 28.8, 27.8, 22.4, 14.3.

MS (EI, 70 eV): m/z (%) = 416 (M^+ , 75), 206 (15), 205 (25), 140 (22), 139 (46), 129 (14), 115 (17), 91 (100), 43 (15).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{28}\text{H}_{32}\text{OS}$: 416.2174; found: 416.2167.

1-Methoxy-4-[(1E)-1-(4-methylphenyl)-2-phenyloct-1-enyl]thio]benzene (9b); Typical Procedure (TP5)

In a dry and argon-flushed Schlenk flask equipped with a septum and a magnetic stirring bar, **6c** (181 mg, 0.4 mmol) was dissolved in THF (0.5 mL). Then, $\text{Pd}(\text{dba})_2$ (6.9 mg, 3 mol%), $\text{P}(o\text{-furyl})_3^{17}$ (5.6 mg, 6 mol%), and *p*-tolylzinc iodide (**8b**; 0.67 M in THF; 0.9 mL, 0.6 mmol) were added at 0 °C and the mixture was warmed to 25 °C and stirred at this temperature for 5 h. The mixture was quenched with sat. NH_4Cl soln (10 mL) and the resulting mixture was extracted with Et_2O (3×10 mL). The combined organic layers were dried (MgSO_4), filtered, and concentrated under reduced pressure. Purification by flash chromatography (pentane + 2 vol% Et_3N) yielded **9b** (90 mg, 54%, *E/Z* 97:3) as a colorless oil.

IR (diamond-ATR, neat): 2923 (m), 1593 (m), 1491 (s), 1462 (m), 1441 (m), 1285 (m), 1243 (vs), 1179 (m), 1172 (m), 1032 (s), 821 (s), 804 (s), 766 (m), 698 (s), 640 cm^{-1} (m).

^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ = 7.14–7.01 (m, 7 H), 6.90–6.87 (m, 2 H), 6.75–6.71 (m, 4 H), 3.63 (s, 3 H), 2.96 (t, J = 7.0 Hz, 2 H), 2.05 (s, 3 H), 1.29–1.19 (m, 8 H), 0.83–0.79 (m, 3 H).

^{13}C NMR (100 MHz, $\text{DMSO}-d_6$): δ = 163.3, 163.3, 151.8, 147.2, 141.6, 140.5, 137.0, 135.6, 134.2, 133.0, 133.0, 131.5, 130.0, 119.6, 60.2, 44.7, 42.7, 36.2, 33.6, 33.0, 27.2, 25.7, 19.1.

MS (EI, 70 eV): m/z (%) = 417 (31), 416 (M^+ , 100), 237 (16), 131 (18), 117 (20), 105 (31), 91 (18).

HRMS (EI): m/z [M] $^+$ calcd for $\text{C}_{28}\text{H}_{32}\text{OS}$: 416.2174; found: 416.2162.

(5E)-6-[(4-Methylphenyl)thio]-5,7-diphenylhept-5-enenitrile (9c)

Prepared according to TP5 from **6d** (105 mg, 0.25 mmol) and BnZnCl (**8a**; 1.19 M in THF; 0.32 mL, 0.38 mmol) (reaction conditions: 25 °C, 6 h). Purification by flash chromatography (pentane– Et_2O , 5:1 + 2 vol% Et_3N) yielded **9c** (73 mg, 76%, *E/Z* 99:1) as a colorless oil.

IR (diamond-ATR, neat): 3025 (w), 2921 (w), 1491 (m), 1452 (m), 1262 (w), 1210 (w), 1180 (w), 1083 (w), 806 (s), 774 (m), 748 (m), 698 (vs), 620 cm⁻¹ (w).

¹H NMR (400 MHz, DMSO-*d*₆): δ = 7.42–7.38 (m, 2 H), 7.33–7.29 (m, 3 H), 7.21–7.10 (m, 7 H), 6.90–6.87 (m, 2 H), 3.31 (s, 2 H), 2.90–2.86 (m, 2 H), 2.48–2.44 (m, 2 H), 2.28 (s, 3 H), 1.58–1.50 (m, 2 H).

¹³C NMR (100 MHz, DMSO-*d*₆): δ = 147.6, 141.1, 139.3, 136.5, 131.2, 130.4, 130.2, 130.1, 129.0, 128.6, 128.5, 128.5, 127.8, 126.4, 120.8, 38.6, 35.8, 24.0, 21.0, 16.4.

MS (EI, 70 eV): *m/z* (%) = 384 (27), 383 (M⁺, 100), 205 (28), 182 (15), 124 (13), 115 (11), 91 (51), 44 (11).

HRMS (EI): *m/z* [M]⁺ calcd for C₂₆H₂₅NS: 383.1708; found: 383.1699.

(5*E*)-6-(4-Methylphenyl)-6-[(4-methylphenyl)thio]-5-phenyl-hex-5-enenitrile (**9d**)

Prepared according to TP5 from **6e** (126 mg, 0.3 mmol) and *p*-tolylzinc iodide (**8b**; 0.67 M in THF; 0.6 mL, 0.4 mmol) (reaction conditions: 25 °C, 5 h). Purification by flash chromatography (pentane–Et₂O, 5:1 + 2 vol% Et₃N) yielded **9d** (94 mg, 82%, *E/Z* 99:1) as a white solid; mp 70–71 °C.

IR (diamond-ATR, neat): 2922 (m), 1491 (m), 1454 (m), 1441 (m), 1019 (m), 820 (s), 804 (s), 766 (s), 753 (m), 701 (vs), 681 (m), 638 cm⁻¹ (m).

¹H NMR (400 MHz, DMSO-*d*₆): δ = 7.18–7.07 (m, 7 H), 6.99–6.94 (m, 4 H), 6.76–6.73 (m, 2 H), 3.08–3.04 (m, 2 H), 2.50–2.47 (m, 2 H), 2.15 (s, 3 H), 2.05 (s, 3 H), 1.59–1.52 (m, 2 H).

¹³C NMR (100 MHz, DMSO-*d*₆): δ = 146.3, 141.3, 136.3, 135.6, 135.5, 132.0, 130.8, 130.4, 129.5, 129.3, 129.1, 128.0, 127.9, 126.7, 120.4, 36.7, 23.9, 20.6, 20.5, 16.0.

MS (EI, 70 eV): *m/z* (%) = 383 (M⁺, 100), 260 (83), 219 (41), 204 (22), 69 (28), 57 (35), 43 (50), 41 (23).

HRMS (EI): *m/z* [M]⁺ calcd for C₂₆H₂₅NS: 383.1708; found: 383.1694.

Acknowledgment

We thank the European Research Council (ERC) for financial support. We also thank W. C. Heraeus (Hanau), Chemetall GmbH (Frankfurt) and BASF AG (Ludwigshafen) for the generous gift of chemicals. E.Z. also thank German Academic Exchange Service (DAAD) for financial support [programs 'Michail Lomonosov – Forschungsstipendien (B)' (No. A0772362); 'Forschungs- und Arbeitsaufenthalte ausländischer Hochschullehrer und Wissenschaftler' (No. A11/044/12)].

References

- Stüdemann, T.; Knochel, P. *Angew. Chem., Int. Ed. Engl.* **1997**, *36*, 93.
- (a) Normant, J. F.; Alexakis, A. *Synthesis* **1981**, 841.
(b) Robertson, D. W.; Katzenellenbogen, J. A.; Hayes, J. R.; Katzenellenbogen, B. S. *J. Med. Chem.* **1982**, *25*, 167.
(c) Harper, M. J. K.; Walpole, A. L. *Nature* **1966**, *212*, 87.
(d) Al-Hassan, M. I. *Synth. Commun.* **1987**, *17*, 1247.
(e) Kamei, T.; Tami, K.; Yoshida, J. *Adv. Synth. Catal.* **2004**, *346*, 1824. (f) Reiser, O. *Angew. Chem. Int. Ed.* **2006**, *45*, 2838. (g) Abramovitch, A.; Marek, I. *Eur. J. Org. Chem.* **2008**, 4924. (h) Miller, R. B.; Al-Hassan, M. I. *J. Org. Chem.* **1985**, *50*, 2121.
- (a) Ogilvie, W. W.; Flynn, A. B. *Chem. Rev.* **2007**, *107*, 4698. (b) Itami, K.; Kamei, T.; Yoshida, J. *J. Am. Chem. Soc.* **2003**, *125*, 14670. (c) Levin, A.; Basheer, A.; Marek, I. *Synlett* **2010**, 329. (d) Das, J. P.; Chechik, H.; Marek, I. *Nat. Chem.* **2009**, *1*, 128. (e) Zhou, C.; Larock, R. C. *Org. Lett.* **2005**, *7*, 259. (f) Alonso, F.; Beletskaya, I. P.; Yus, M. *Chem. Rev.* **2004**, *104*, 3079. (g) Gerard, J.; Hevesi, L. *Tetrahedron* **2004**, *60*, 367. (h) Yasui, H.; Yorimitsu, H.; Oshima, K. *Bull. Chem. Soc. Jpn.* **2008**, *81*, 373. (i) Yasui, H.; Yorimitsu, H.; Oshima, K. *Chem. Lett.* **2007**, *36*, 32. (j) Kanemura, S.; Kondoh, A.; Yorimitsu, H.; Oshima, K. *Org. Lett.* **2007**, *9*, 2031. (k) Creton, I.; Marek, I.; Brasseur, D.; Jestin, J.-L.; Normant, J.-F. *Tetrahedron Lett.* **1994**, *35*, 6873. (l) Chechik-Lankin, H.; Marek, I. *Org. Lett.* **2003**, *5*, 5087.
- (a) Murakami, K.; Yorimitsu, H.; Oshima, K. *Chem. Eur. J.* **2010**, *16*, 7688. (b) Murakami, K.; Yorimitsu, H.; Oshima, K. *Org. Lett.* **2009**, *11*, 2373.
- (a) Knochel, P.; Singer, R. D. *Chem. Rev.* **1993**, *93*, 2117. (b) Knochel, P.; Almena Perea, J. J.; Jones, P. *Tetrahedron* **1998**, *54*, 8275.
- (a) Rao, S. A.; Knochel, P. *J. Am. Chem. Soc.* **1991**, *113*, 5735. (b) Rao, S. A.; Knochel, P. *J. Am. Chem. Soc.* **1992**, *114*, 7579.
- (Di)organozinc reagents were prepared according to the literature: (a) Metzger, A.; Schade, M. A.; Manolikakes, G.; Knochel, P. *Chem. Asian J.* **2008**, *3*, 1678. (b) Metzger, A.; Schade, M. A.; Knochel, P. *Org. Lett.* **2008**, *10*, 1107. (c) Piller, F. M.; Metzger, A.; Schade, M. A.; Haag, B. A.; Gavryushin, A.; Knochel, P. *Chem. Eur. J.* **2009**, *15*, 7192. (d) Bernhardt, S.; Metzger, A.; Knochel, P. *Synthesis* **2010**, 3802. (e) Metzger, A.; Bernhardt, S.; Manolikakes, G.; Knochel, P. *Angew. Chem. Int. Ed.* **2010**, *49*, 4665.
- (a) Knochel, P.; Yeh, M. C. P.; Berk, S. C.; Talbert, J. J. *J. Org. Chem.* **1988**, *53*, 2390. (b) Dübner, F.; Knochel, P. *Angew. Chem. Int. Ed.* **1999**, *38*, 379.
- Krasovskiy, A.; Knochel, P. *Angew. Chem. Int. Ed.* **2004**, *43*, 3333.
- Alexakis, A.; Commerçon, A.; Villiéras, J.; Normant, J. F. *Tetrahedron Lett.* **1976**, *17*, 2313.
- (a) Rambaud, M.; Villiéras, J. *Synthesis* **1984**, 406. (b) Villiéras, J.; Rambaud, M. *Org. Synth.* **1988**, *66*, 220.
- (a) Nakamura, E.; Matsuzawa, S.; Horiguchi, Y.; Kuwajima, I. *Tetrahedron Lett.* **1986**, *27*, 4029. (b) Bourgain-Commerçon, M.; Foulon, J.-P.; Normant, J. F. *J. Organomet. Chem.* **1982**, *228*, 321.
- Metzger, A.; Piller, F. M.; Knochel, P. *Chem. Commun.* **2008**, 5824.
- Stüdemann, T.; Ibrahim-Ouali, M.; Knochel, P. *Tetrahedron* **1998**, *54*, 1299.
- (a) Negishi, E. *Acc. Chem. Res.* **1982**, *15*, 340. (b) Negishi, E.; Valente, L. F.; Kobayashi, M. *J. Am. Chem. Soc.* **1980**, *102*, 3298. (c) Negishi, E.; King, A. O.; Okukado, N. *J. Org. Chem.* **1977**, *42*, 1821.
- Walker, S. D.; Barder, T. E.; Martinelli, J. R.; Buchwald, S. L. *Angew. Chem. Int. Ed.* **2004**, *43*, 1871.
- (a) Farina, V.; Krishnan, B. *J. Am. Chem. Soc.* **1991**, *113*, 9585. (b) Farina, V.; Kapadia, S.; Krishnan, B.; Wang, C.; Liebeskind, L. S. *J. Org. Chem.* **1994**, *59*, 5905. (c) Klement, I.; Rottländer, M.; Tucker, C. E.; Majid, T. N.; Knochel, P.; Venegas, P.; Cahiez, G. *Tetrahedron* **1996**, *52*, 7201.
- Melzig, L.; Metzger, A.; Knochel, P. *Chem. Eur. J.* **2011**, *17*, 2948.